BT503- QUIZ-01 (87 MCQs) 23-08-2020

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1- ------- are well suited to large scale treatment where speed is not of the essence. Extensive technologies
2- Cropping of heavy metal accumulator plants occur in ------ technology. Extensive technologies
3- The design of sludge treatment and final disposal stages is based on which of the following. Both
4- In situ technique is suitable for -------- contamination. Widespread
5- Among dewatering process the process which is least used is called -------. Sludge lagoons
6- Which of the following sludge management stage accomplished through the removal of the biodegradable organic matter of the sludge? stabilization
7- Capillary water maintains itself adsorbed in which of the following phase. Solid
8- Most frequently used unit for concentration of sludge is ------. %
9- Which of following process does not introduce chemical products into the liquid? Membranes
10- ------- technique is the best suited to instances of relatively localized pollution within a site typically in hotspot of medium to relatively high concentration. Ex-situ
11- Soil washing and thermal treatment are examples of ------ technologies. Intensive technologies
12- Which of the following method affects the reproduction of pathogenic agents? UV radiation
13- Facultative ponds require how much time for sludge removal from liquid phase. Decades
14- All of these are artificial process of pathogen removal except. Land treatment
15- ------- following chemicals are added in the sludge preparation process. Both a & c
16- A sludge with a level of dry solids of 2% has water content of -------. 98%
17- A sludge with a solids load of 120 kgSS/d and a solids concentration of 2.0% will have a flow of. 6.0 m^3/d
18- A very effective method for removal of disease causing organisms is -----. Ozonisation

19- The main characteristics of ------ methods are that the soil is removed from where it originally lay for treatment. Ex-situ
20- Which of the following pathogen removal system does not need chemicals? Maturation ponds
21- Which of the following is highly recommended system for pathogen removal? Maturation ponds
22- Disposal of sludge in the mining areas is known as -----. Land reclamation
23- The process in which decomposition of aerobic organic matter is achieved through controlled conditions of temperature, water content and nutrients is known as -----. Composting
24- ------- is normally uses high temperature to fuse the contaminated materials. Vitrification
25- In a dewatering process the dry solid level in the sed sludge of filter press. 20-30%
26- For preliminary waste water treatment all basic equipment are required except one. Aerator
27- Transformation or mineralization of contaminants to less toxic and more mobile, forms is known as ____ remediation method. Biological
28- ____process is an appropriated method for sludge stabilization in nurseries, horticulture and landscaping. Composting
29- How much dry solid content is found in fluid sludge? 0%-25%
30- Which basic equipment is required for high rate pond waste water treatment process? Rotors for liquid movement
31- Water content influences which properties of the sludge? Mechanical
32- _____process is an encapsulation of contaminants within a monolithic solid of high structural integrity, with or without associated chemical fixation. Vitrification
33- Adsorbed water in the sludge can be removed by? Flocculating agents
34- Sludge drying lagoons are used for. All
35- ------the following technologies have high initiation, running and support costs. Intensive technologies
36- The sludge production is a function of the waste water treatment system used for the_____. Liquid phase
37- _____is a land disposal process, in which the organic substrate is biologically degraded in the upper layer of the soil and the inorganic fraction is transformed or fixed into this layer. Land Farming
38- How much water content is found in fluid sludge? 100%-75%
39- ------technologies have smaller resource requirement and lower initiation, running and support costs. Extensive technologies

40- For sludge stabilization, chemical stabilization is accompanied by ------of an organic matter. Oxidation

41- Toxic by-products can be generated in which of the following system? Chlorination

42- Thermal decomposition process by oxidation is known as____. Incineration

43- Type of centrifuges mostly used in the dewatering process are called--------. Horizontal-Shaft Centrifuge

44- --------is an example of thermal remediation method. Gasification

45- For which of the following equipment depends upon the post-treatment process used. UASB reactor + post-treatment

46- Stages of sludge processing involve? _ALL

47- _____ is very effective agent for the removal of pathogens. ___Ozone

48- of the liquid sludge range between? ____ 1.02 and 1.03

49- If the sludge is to be incinerated then the following --------process is not necessary. _____Disinfection

50- Water in the sludge can be classified into how many classes?___ 4

51- In primary treatment system, primary sludge takes how much time for sludge removal from liquid phase? months

52- Lime stabilization is used to treat_____ sludge. ___All

53- The concentration of solids in sludge is expressed in the form of? _____ Dry solids

54- - metallic coagulants is/are used in conditioning of sludge._____ both A&B

55- Principle mechanism of influence in the dewatering process is called -------. _____Evaporation

56- Free water in the sludge can be removed by….gravity.

57- Pasteurisation ……..involves the heating of sludge at 70°C for 30 mints.

58- Normal temperature required in thermopilic aerobic digestion process is……50-60°C

59- For primary waste water treatment, all basic equipments are required except….grit chamber.

60- The period required for carry out composition through in vessel biological reactors is normally……14-21days

61- Which one of the following sludge management stage includes the removal of organic matter? Stabilization

62- Criteria for management of waste water treatment include……all

63- …..agent/s responsible for land contamination……..asbestos work

64- --------- is the main water removal mechanism in the sludge dewatering……. Percolation.

65- Time required for the accumulation of discard sludge from WWTPs…

3-5years

66- The density of the sludge during most of its processing is very close to…. Water.
67- In sludge processing stages specific gravity is close to -----. 0.1
68- Density of the liquid sludge range between ------. 1.02-1.03
69- Density of sludge in processing is close to --------.1000kg/m3
70- In extensive technologies precipitation of metal sulphides take place under ----- condition. Anaerobic
71- Removal of organic matter in sludge management is known as --------. Mass reduction
72- Which method are costly for removal of disease causing organisms. Artificial
73- The pH of the medium will be ----- when time is addad to the sludge in the sludge disinfection process. Basic
74- In UASB + polishing pond dry solid level ranges between. 3-4%
75- -------- is the main natural water removal mechanism in the sludge dewatering. Percolation
76- A process in sludge management for preparation for dewatering is known as ------. Conditioning
77- ------ technologies impose less damage to soil quality. Extensive technologies.
78- ------ is used for thickening of sludge. Both b& c
79- A common way in which all form of remediation are often characterized is as -----. Both a & b
80- In the process of dissolved air flotation release of oxygen in the form of bubbles is known as --------. Depressurization
81- Virtification uses ----- temperature to fuse contaminated materials. High
82- Gamma radiations are used for ----------. Disinfection
83- Which of following technologies key factors are a fast response and low treatment time are key factors which pertains to ----------. Intensive technologies
84- ------ dewatering can be done by natural or mechanical processes. Sludge
85- Sludge treatment stages are ----------. 6
86- Water in the sludge can be divided into ------- classes. 4
87- Thickening in sludge management is removal of --------. Water

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